

Email Classification

RedRock¹, a financial services company, processes thousands of client emails every day. They need a way of automatically classifying incoming emails into appropriate business categories. Based on the categories, they will forward each email to the correct department to ensure regulatory compliance.

You are given a folder of sample emails from January 2025 to August 2025, as well as a .csv file with some manually classified examples. Your task is to build a classification model that will assign the remaining emails to one of 5 business categories.

Your solution should consist of the following components:

1. Ingestion
 - Read from the provided folder and .csv file.
 - Extract and preprocess the email body and metadata (subject line, etc.).
2. Modelling
 - Use an appropriate machine learning based approach (LLMs or other NLP techniques) to classify emails into the 5 business categories: Account Management, Investment Advisory, Loan Processing, Insurance Claims and Other.
 - Your model should output both a predicted category and a confidence score for each email.
3. Simple reporting
 - Write your results to a .csv file with the following column headings: email_id, predicted_category and confidence_score.

The financial industry is heavily regulated and incorrect forwarding of emails can result in delays and even penalties. Explain how you would measure the success of your method. Also explain what additional features or data sources could increase classification accuracy to improve regulatory compliance.

The above steps should run end-to-end (.csv to results). Your solution must be written using only open source tools (e.g. python, R, ...) so we can run your solution and check your results.

¹ A fictitious financial administrator